

● Part A: Conceptual Understanding (Answers)

1. What are Supervised Learning Algorithms?

Supervised learning algorithms are machine learning methods that learn from **labeled data**, where both input and output are known. The model learns the relationship between features (X) and target (y) to make predictions on new data. Examples include regression and classification algorithms.

2. Difference between Regression Algorithms vs Classification Algorithms

Regression algorithms predict **continuous values** (e.g., house price), while classification algorithms predict **categorical labels** (e.g., spam or not spam). Regression outputs numbers, whereas classification outputs classes or categories.

3. Explain Simple Linear Regression

Simple Linear Regression is a technique used to model the relationship between **one independent variable (X)** and a **dependent variable (y)** using a straight line.

where m is the slope and b is the intercept.

4. Assumptions of Linear Regression

Linear Regression assumes:

- A **linear relationship** between input and output
 - **Independence** of errors
 - **Constant variance (homoscedasticity)**
 - Errors are **normally distributed**
 - No **multicollinearity** (in multiple regression)
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5. What is Bias-Variance Trade-Off?

The bias-variance trade-off refers to balancing **underfitting (high bias)** and **overfitting (high variance)**. A good model minimizes both to achieve optimal performance on unseen data.

6. Explain Overfitting and Underfitting with Examples

Overfitting occurs when a model learns noise from training data and performs poorly on new data (e.g., very complex polynomial).

Underfitting occurs when a model is too simple to capture patterns (e.g., fitting a straight line to nonlinear data).